

A PROJECT REPORT

On

SmartDrive – 3D Self-Driving Car Simulator

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CERTIFICATE

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CONTENT

Certificate	1
Acknowledgement	2
List of Figure/ Table.....	5
Abstract	6
1. Preface	
1.1. Introduction	7-8
1.2. Motivation	8-9
1.3. Contribution	9-10
2. Problem Statement	11
3. Literature Survey	
3.1. Introduction	12
3.2. Review of Existing Systems	13-16
3.2.1 Rule Based Autonomous Driving	13
3.2.2 Machine Learning-Based Autonomous Driving	13-14
3.2.3 Deep Learning for Autonomous Vehicles	14
3.2.4 Reinforcement Learning in Autonomous Navigation	14-15
3.2.5 Sensor-Based Perception System	15
3.2.6 Path Planning Algorithms	15-16
3.2.7 Autonomous Driving Simulation Platfrom	16
3.3. Comparative Analysis of Existing Systems	17
3.4. Research Gap	17-18
3.5. Summary	18
4. Flow of the Work	19
4.1. Phase 1: System Initialization	20
4.2. Phase 2: Route Generation and Navigation Preparation	20-21
4.3. Phase 3: Autonomous Driving and Environment Interaction	21-22
4.3.1 Vehicle Dynamic Update	21
4.3.2 Environmental Perception	21
4.3.3 Autonomous Decision Making	21-22

4.3.4 Traffic and Pedestrian Management	22
4.4. Phase 4: Telemetry Collection and Performance Analysis	22-23
5. Use Case Diagram	24-28
6. Data Flow Diagram	27
7. Proposed System Algorithm	
7.1. Introduction	28
7.2. Assumptions	28-29
7.3 Overall Algorithm.....	29-30
7.4 Pseudocode	30
7.5 Working of the Algorithm	30-31
7.6. Backend Decision Making.....	31
7.7. Frontend Vehicle Control... ..	31
7.8. Mathematical Mode	32
7.9. Advantages of the Proposed Algorithm.....	32
8. Result and Screenshot.....	
8.1. Result	35-38
8.1.1 Experimental Setup	34
8.1.2 Experimental Result	35
8.1.3 Comparative Analysis	36
8.1.4 Performance Analysis	37
8.2. System Screenshot	38-39
9. Conclusion and Future Work	40
10. Reference	41

LIST OF TABLES

Table No.	Table Name	Page No.
Table 1	Comparative Analysis of Existing Systems	17
Table 2	Description of Use Case	25
Table 3	Simulation Parameters	34
Table 4	Experimental Results	35
Table 5	Comparative Analysis of Traditional Simulator and SmartDrive Simulator	36
Table 6	Functional Testing Results	37
Table 7	Performance Summary	36

LIST OF FIGURES

Figure No.	Figure Name	Page No.
Figure 1	Flow of the Work	19
Figure 2	Use Case Diagram	26
Figure 3	Level 0 Data Flow Diagram	27
Figure 4	Level 1 Data Flow Diagram	27
Figure 5	Autopilot Efficiency and Safety under Different	35
Figure 6	Home Screen of SmartDrive Simulator	38
Figure 7	Route Selection and Autopilot Activation	39

Abstract

Autonomous driving is one of the most significant applications of Artificial Intelligence (AI), computer vision, robotics, and intelligent transportation systems. Developing a safe and reliable self-driving vehicle requires accurate perception of the surrounding environment, efficient route planning, intelligent decision-making, and realistic vehicle control. This project presents SmartDrive, a 3D self-driving car simulator developed to demonstrate autonomous navigation in a controlled virtual environment using modern web technologies and artificial intelligence techniques [1] [2].

The simulator is built using Three.js for real-time 3D visualization, Vite as the frontend development environment, and Python Flask as the backend server for route planning and telemetry management. The system models an urban road network consisting of intersections, curved roads, parking bays, traffic signals, zebra crossings, pedestrians, and dynamic traffic vehicles. An A* (A-Star) pathfinding algorithm is employed to compute the shortest route between selected locations. The autonomous vehicle follows the generated route using a Pure Pursuit-based steering controller while continuously adjusting its speed according to road conditions [2] [7].

To improve driving safety, the vehicle is equipped with a simulated Lidar sensing system based on ray casting. The sensor continuously detects nearby vehicles, pedestrians, and road boundaries, enabling collision avoidance, traffic signal compliance, pedestrian crossing detection, and adaptive speed control. The simulator also incorporates realistic vehicle dynamics, multiple camera modes, telemetry logging, graphical performance analysis, and configurable environmental settings such as traffic density and day–night lighting conditions. The proposed system demonstrates that integrating intelligent path planning, sensor-based perception, realistic vehicle physics, and rule-based autonomous decision-making can produce smooth, reliable, and safe autonomous navigation within a simulated environment. The project provides an effective educational and research platform for studying autonomous driving algorithms while serving as a scalable foundation for future enhancements, including computer vision, deep reinforcement learning, cloud-based vehicle communication, and advanced intelligent transportation systems [8] [9].

1. Preface

1.1 Introduction

The SmartDrive – 3D Self-Driving Car Simulator is a web-based autonomous driving simulation platform developed to demonstrate the fundamental concepts of intelligent vehicle navigation within a realistic three-dimensional urban environment [1] [2]. The project integrates Artificial Intelligence, path-planning algorithms, sensor-based perception, vehicle dynamics, traffic management, and real-time visualization into a single modular system. Its primary objective is to provide a safe, interactive, and cost-effective platform for understanding and evaluating autonomous driving technologies without requiring expensive physical hardware or real-world vehicle testing.

The simulator is developed using Three.js [4] for real-time 3D graphics rendering, Vite as the frontend development environment, and Python Flask [5] as the backend framework responsible for route generation, telemetry processing, and communication between different system modules. The frontend provides an immersive driving environment containing roads, intersections, traffic signals, pedestrian crossings, moving vehicles, and pedestrians, while the backend performs computational tasks required for intelligent navigation and system management.

The autonomous vehicle navigates through the virtual environment using the A* (A-Star) path-planning algorithm, which computes the shortest route between the selected source and destination [2] [7]. During navigation, a simulated LiDAR sensor implemented through ray-casting continuously scans the surrounding environment to detect nearby vehicles, pedestrians, road boundaries, and obstacles [8]. Based on this information, the autonomous controller performs intelligent steering, adaptive speed control, obstacle avoidance, traffic signal compliance, and safe navigation.

In addition to autonomous navigation, the simulator incorporates realistic vehicle physics to model acceleration, braking, steering, turning, and friction, producing smooth and natural vehicle movement. Dynamic traffic vehicles and pedestrians create realistic urban driving

scenarios, allowing the autonomous controller to respond intelligently to changing traffic conditions. Throughout the simulation, operational data such as vehicle speed, travelled distance, steering angle, braking events, collision statistics, and route completion status are continuously recorded through the telemetry system for performance evaluation and future analysis.

The project has been designed with a modular architecture that allows individual components such as the road network, sensor system, autonomous controller, backend services, and visualization layer to operate independently while communicating efficiently. This modularity simplifies maintenance and enables future integration of advanced technologies including Computer Vision, Deep Learning, Reinforcement Learning, and Vehicle-to-Everything (V2X) communication. Consequently, SmartDrive serves as both an educational platform and a research-oriented framework for studying intelligent transportation systems and autonomous driving technologies.

1.2 Motivation

The rapid advancement of **Artificial Intelligence (AI)**, **Machine Learning (ML)**, and **Intelligent Transportation Systems (ITS)** has significantly accelerated the development of autonomous driving technologies. Self-driving vehicles are being developed to improve road safety, reduce traffic congestion, and enhance transportation efficiency. These advancements motivated the development of the **SmartDrive – 3D Self-Driving Car Simulator** as an educational and research-oriented platform.

One of the primary motivations behind this project is the need for a safe and cost-effective environment to develop and evaluate autonomous driving algorithms. Real-world testing requires expensive hardware such as LiDAR sensors, cameras, GPS modules, and high-performance computing devices, while also involving safety risks and regulatory challenges. A simulation-based platform provides a practical alternative by enabling autonomous driving experiments in a controlled virtual environment.

Another motivation is the lack of lightweight educational simulators that integrate essential autonomous driving components into a single system. Many existing simulators are either complex and resource-intensive or demonstrate only limited functionalities. This project

combines intelligent route planning, LiDAR-based obstacle detection, realistic vehicle physics, traffic management, and telemetry analysis within one modular platform.

The project also bridges the gap between theoretical learning and practical implementation by integrating concepts from Artificial Intelligence, Machine Learning, Computer Graphics, Software Engineering, and Intelligent Transportation Systems into a real-time simulation. Furthermore, its modular architecture allows future integration of advanced technologies such as Computer Vision, Deep Learning, Reinforcement Learning, and Vehicle-to-Everything (V2X) communication.

The **SmartDrive – 3D Self-Driving Car Simulator** therefore serves as a realistic, scalable, and cost-effective platform for learning, research, and experimentation in the field of autonomous driving.

1.3 Contribution

The major contributions of the proposed **SmartDrive – 3D Self-Driving Car Simulator** are summarized as follows:

- Developed a realistic **3D autonomous driving simulation platform** using Three.js and Python Flask.
- Designed a modular software architecture that separates frontend visualization, backend services, autonomous control, and telemetry processing for improved scalability and maintainability.
- Implemented intelligent route planning using the A* (A-Star) algorithm to generate optimal navigation paths between selected locations.
- Developed a **ray-casting-based LiDAR simulation** for real-time detection of vehicles, pedestrians, and road obstacles.
- Implemented realistic vehicle dynamics, including steering, acceleration, braking, turning, and friction, to simulate natural driving behaviour.
- Integrated dynamic traffic vehicles, traffic signals, and pedestrian movement to create realistic urban driving scenarios.

- Developed an autonomous driving controller capable of performing intelligent navigation, obstacle avoidance, adaptive speed control, and traffic rule compliance.
- Implemented a telemetry system for recording operational parameters such as vehicle speed, travelled distance, steering angle, braking events, collision statistics, and route completion.
- Built a lightweight web-based platform that eliminates the need for expensive autonomous vehicle hardware while providing an effective environment for experimentation and learning.

2. Problem Statement

The rapid advancement of autonomous vehicle technology has transformed modern transportation by integrating Artificial Intelligence (AI), Machine Learning (ML), sensor technology, and intelligent control systems to improve road safety and driving efficiency. However, developing a reliable autonomous driving system that can safely operate under dynamic traffic conditions remains a significant challenge.

An autonomous vehicle must continuously detect roads, vehicles, pedestrians, traffic signals, and obstacles while simultaneously planning the optimal route and controlling steering, acceleration, and braking. These tasks must be performed in real time with high accuracy to ensure safe and efficient navigation. Testing such systems on real roads requires expensive hardware, including LiDAR sensors, cameras, GPS modules, and high-performance computing devices, making the development process costly and complex. Moreover, real-world testing involves safety risks and makes it difficult to reproduce identical traffic scenarios for evaluation.

Simulation provides a safe and cost-effective alternative for developing and testing autonomous driving systems. Although several simulation platforms are available, many require powerful hardware and complex software configurations, while lightweight simulators often lack essential features such as intelligent route planning, realistic vehicle physics, obstacle detection, traffic management, pedestrian interaction, and performance analysis. Consequently, students and researchers often do not have access to a single platform that integrates all major autonomous driving components.

To address these challenges, the proposed **SmartDrive – 3D Self-Driving Car Simulator** has been developed as a lightweight and modular simulation platform. The system integrates A* path planning, LiDAR-based obstacle detection, realistic vehicle physics, traffic management, pedestrian interaction, and telemetry analysis within a three-dimensional virtual environment. This enables users to safely develop, test, and evaluate autonomous driving algorithms while providing an effective platform for education and research.

3. Literature Survey

3.1 Introduction

The rapid evolution of autonomous driving technology has significantly transformed the field of intelligent transportation systems over the past two decades. Advances in Artificial Intelligence (AI), Machine Learning (ML), Robotics, Computer Vision, and Sensor Technology have enabled vehicles to perceive their surroundings, make intelligent decisions, and navigate safely with minimal human intervention. Autonomous vehicles have the potential to reduce road accidents, improve traffic flow, minimize fuel consumption, and provide safer transportation by reducing the dependency on human drivers [1] [14].

Developing a reliable autonomous driving system, however, is a multidisciplinary challenge. It requires the integration of environmental perception, path planning, vehicle control, obstacle detection, traffic management, and real-time decision-making. Researchers around the world have proposed numerous techniques to address these challenges, ranging from traditional rule-based control systems to advanced deep learning and reinforcement learning approaches [3] [15].

In addition to algorithm development, simulation has become an indispensable part of autonomous driving research. Virtual simulation environments provide a safe, repeatable, and cost-effective platform for testing autonomous navigation algorithms without exposing expensive hardware or human lives to unnecessary risks. Numerous simulation platforms have been developed for this purpose, each providing different levels of realism, flexibility, and computational complexity [8] [9].

This chapter reviews the major research contributions related to autonomous driving technologies, vehicle perception systems, path-planning algorithms, intelligent control strategies, and simulation platforms. It also identifies the limitations of existing approaches and

establishes the motivation for developing the proposed **SmartDrive – 3D Self-Driving Car Simulator** .

3.2 Review of Existing Systems

Several autonomous driving systems have been developed using different technologies and Artificial Intelligence techniques. Each approach focuses on solving specific challenges associated with autonomous navigation, including environmental perception, vehicle control, obstacle avoidance, and intelligent decision-making. The major approaches are discussed below.

3.2.1 Rule-Based Autonomous Driving Systems

The earliest autonomous driving systems relied primarily on predefined logical rules and mathematical models. In these systems, engineers manually programmed the vehicle to perform specific actions under predefined traffic conditions. Common operations included lane keeping, speed regulation, obstacle avoidance, and traffic signal compliance.

Rule-based systems are highly deterministic, making their behavior predictable and easy to verify. Since every decision follows explicitly defined rules, these systems are suitable for safety-critical applications where transparency is essential. However, they cannot adapt to unfamiliar situations or learn from previous experiences. As traffic environments become increasingly dynamic and complex, maintaining a large number of manually programmed rules becomes inefficient [1].

3.2.2 Machine Learning-Based Autonomous Driving

Machine Learning introduced a data-driven approach to autonomous navigation by enabling vehicles to learn driving behavior from historical datasets rather than relying entirely on manually programmed rules.

Supervised learning algorithms such as Decision Trees, Random Forest, Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Gradient Boosting have been widely

used for predicting steering angles, recognizing traffic signs, detecting vehicles, and classifying road conditions.

Compared with traditional rule-based systems, machine learning provides better adaptability and prediction accuracy. However, its performance depends heavily on the quality and diversity of training datasets. Models often struggle when encountering driving situations that were not represented during training [13].

3.2.3 Deep Learning for Autonomous Vehicles

The introduction of Deep Learning significantly improved autonomous vehicle perception by enabling computers to automatically extract complex features from images and sensor data.

Convolutional Neural Networks (CNNs) have become the dominant architecture for image-based perception tasks, including lane detection, traffic sign recognition, pedestrian detection, object classification, and semantic segmentation. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have also been applied for trajectory prediction and sequential decision-making.

Although deep learning achieves impressive accuracy, it requires large datasets, powerful GPU hardware, and extensive training time. The lack of explainability and high computational requirements remain important limitations for real-time autonomous driving applications [3].

3.2.4 Reinforcement Learning in Autonomous Navigation

Reinforcement Learning (RL) enables autonomous vehicles to learn driving strategies through continuous interaction with their environment. Instead of relying on labeled datasets, the vehicle receives rewards for desirable actions and penalties for undesirable behavior, gradually improving its navigation policy over time.

Algorithms such as Q-Learning, Deep Q Networks (DQN), Proximal Policy Optimization (PPO), and Actor-Critic methods have demonstrated promising results in lane keeping, obstacle avoidance, adaptive cruise control, and autonomous parking.

Despite its adaptability, reinforcement learning requires extensive training episodes and carefully designed reward functions. During the learning process, the agent may perform unsafe actions, making simulation an essential environment for RL-based autonomous driving research [16].

3.2.5 Sensor-Based Perception Systems

Reliable environmental perception is fundamental to autonomous driving. Modern autonomous vehicles employ multiple sensing technologies to understand their surroundings accurately.

The most commonly used sensors include:

- LiDAR
- Cameras
- Radar
- Ultrasonic Sensors
- GPS
- Inertial Measurement Units (IMU)

LiDAR provides accurate distance measurements through laser scanning, while cameras capture detailed visual information for object recognition and lane detection. Radar performs reliably under adverse weather conditions, and ultrasonic sensors assist in short-range obstacle detection.

Many autonomous driving systems combine multiple sensors through sensor fusion techniques to improve robustness and perception accuracy [8].

3.2.6 Path Planning Algorithms

Path planning is responsible for generating the safest and shortest route between a starting point and a destination.

Several algorithms have been proposed, including:

- Dijkstra Algorithm

- A* (A-Star) Algorithm [2], [7]
- Rapidly Exploring Random Trees (RRT)
- Probabilistic Road Maps (PRM)
- Hybrid A*

Among these approaches, the A* algorithm remains one of the most widely adopted techniques because of its computational efficiency, heuristic-based optimization, and ability to generate near-optimal paths within structured road networks.

The proposed SmartDrive simulator also utilizes the A* algorithm for intelligent route planning.

3.2.7 Autonomous Driving Simulation Platforms

Simulation platforms provide researchers with safe environments for developing and evaluating autonomous driving algorithms.

Popular simulation environments include:

- CARLA [8]
- Microsoft AirSim [9]
- LGSVL Simulator
- Gazebo
- Webots [10]

These platforms support realistic traffic environments, weather conditions, sensor simulation, and vehicle dynamics. However, they often require high-end computing resources and involve complex software configurations, limiting their accessibility for educational institutions.

3.3 Comparative Analysis of Existing Systems

Existing System	Advantages	Limitations
Rule-Based Systems	Simple, deterministic, easy to debug	Cannot adapt to new situations
Machine Learning Systems	Learns driving behavior from data	Requires large labeled datasets
Deep Learning Systems	Excellent perception accuracy	High computational cost
Reinforcement Learning Systems	Adaptive decision-making	Long training time and unstable early learning
Commercial Simulators	Highly realistic environments	Expensive hardware and complex setup

Table 1: Comparative Analysis of Existing Systems

3.4 Research Gap

The literature survey reveals that although substantial progress has been made in autonomous driving research, several limitations remain unresolved.

Most rule-based systems fail to adapt to dynamic traffic environments, whereas machine learning models depend heavily on training datasets. Deep learning techniques require powerful hardware and extensive computational resources, making them difficult to deploy in lightweight educational environments. Reinforcement learning provides adaptability but requires long training durations and carefully designed reward mechanisms.

Similarly, existing autonomous driving simulators either focus on industrial-level realism or simplify the simulation to the extent that important autonomous driving components are omitted. Many educational simulators lack integrated road networks, intelligent path planning, realistic vehicle dynamics, pedestrian interaction, telemetry analysis, and modular software architecture.

These limitations highlight the need for a comprehensive yet lightweight autonomous driving simulator that integrates intelligent navigation, realistic vehicle behavior, sensor-based perception, traffic management, and performance analysis within a single educational platform [8], [9], [10].

3.5 Summary

This chapter reviewed the major research contributions in autonomous driving, including rule-based systems, machine learning, deep learning, reinforcement learning, sensor technologies, path-planning algorithms, and simulation platforms. The review indicates that no single technique is capable of addressing every challenge associated with autonomous driving. Consequently, modern autonomous driving systems increasingly integrate multiple technologies to achieve reliable and safe navigation.

Based on the identified research gaps, the proposed **SmartDrive – 3D Self-Driving Car Simulator** combines A* path planning, ray-casting-based LiDAR perception, realistic vehicle physics, autonomous decision-making, dynamic traffic management, telemetry analysis, and a modular client-server architecture to provide a practical, scalable, and educational platform for autonomous driving research and development.

4. Flow of the Work

The execution flow of the **SmartDrive – 3D Self-Driving Car Simulator** follows a structured sequence consisting of system initialization, route generation, autonomous navigation, and telemetry analysis. The simulator operates using a modular client-server architecture, where the frontend manages visualization and vehicle interaction, while the Flask backend performs route computation, telemetry processing, and system coordination. The overall execution lifecycle is divided into the following phases.

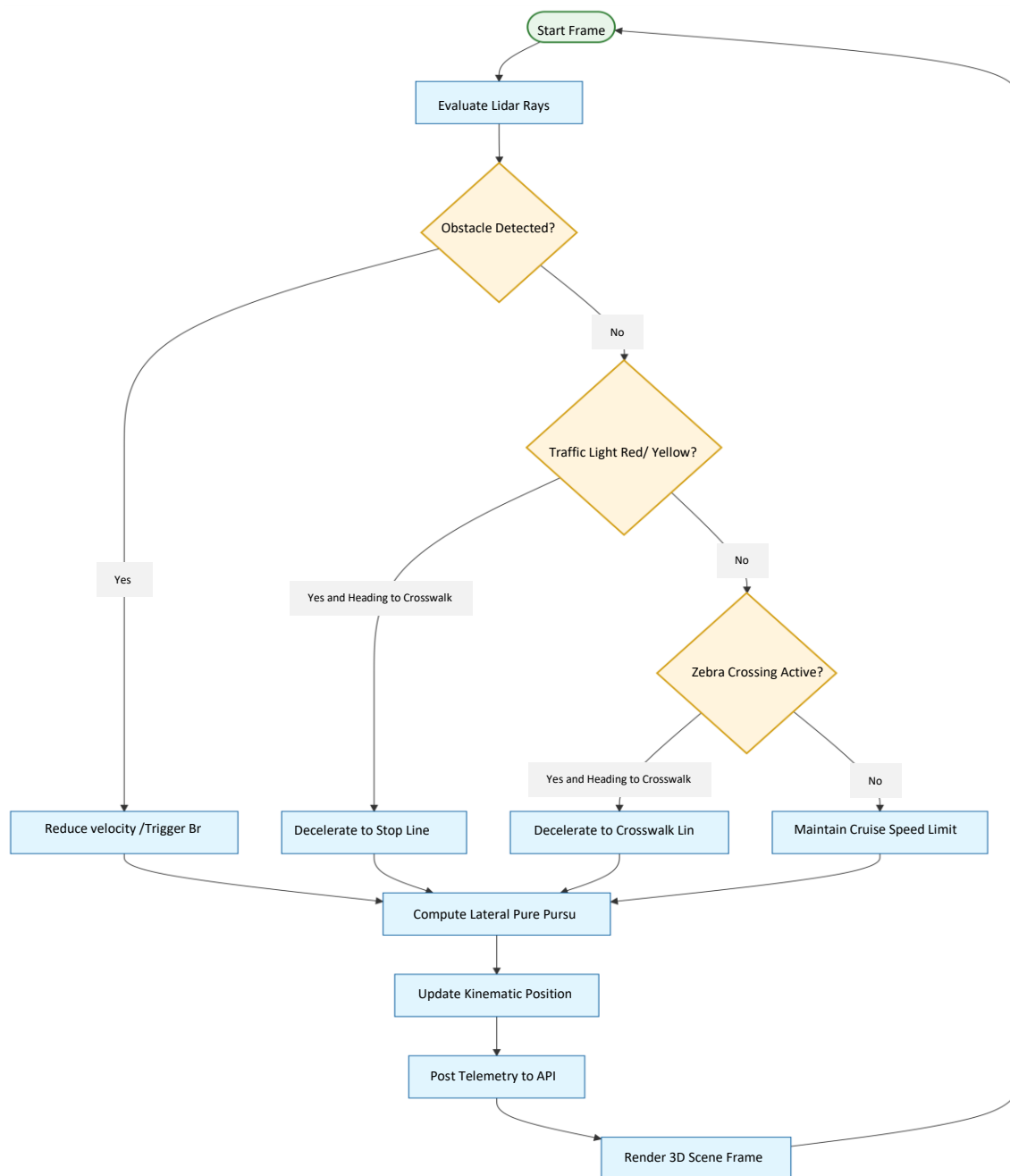


Fig1: Flow of the Work

4.1. Phase 1: System Initialization

The execution begins when the user launches the SmartDrive simulator. During this phase, both the frontend and backend components are initialized to prepare the simulation environment.

The **Python Flask backend** starts and initializes the application services, including the road network, navigation APIs, telemetry services, and configuration parameters required for autonomous navigation. Simultaneously, the **Vite development server** loads the frontend application developed using **Three.js**.

After successful initialization, the frontend establishes communication with the backend and requests the required road network information. Based on the received data, the simulator constructs the complete three-dimensional driving environment, including road segments, lane markings, intersections, parking bays, zebra crossings, traffic signals, environmental objects, and lighting.

Once the virtual environment has been generated, the autonomous vehicle is placed at the selected starting location while dynamic traffic vehicles and pedestrians are initialized throughout the road network to simulate realistic urban traffic conditions.

4.2 Phase 2: Route Generation and Navigation Preparation

After the simulation environment has been created, the user selects the desired **source** and **destination** locations through the graphical user interface.

The selected information is transmitted to the Flask backend, where the **A*** (A-Star) path-planning algorithm computes the shortest and most efficient route across the road network.

The generated sequence of navigation nodes is returned to the frontend. These nodes are then converted into a smooth driving trajectory consisting of multiple waypoints that the autonomous vehicle follows during navigation.

Before autonomous movement begins, the navigation controller initializes the required driving parameters, including steering angle, target waypoint, vehicle orientation, and initial speed. This preparation ensures smooth vehicle movement throughout the journey.

4.3. Phase 3: Autonomous Driving and Environment Interaction

Once the navigation path has been generated, the simulator enters its continuous real-time execution loop. During every simulation frame, multiple autonomous driving components operate simultaneously to ensure safe and efficient navigation.

4.3.1 Vehicle Dynamics Update

The vehicle physics engine continuously updates the position, velocity, acceleration, steering angle, and orientation of the autonomous vehicle using realistic vehicle movement equations. These calculations ensure smooth acceleration, braking, turning, and lane-following behaviour.

4.3.2 Environmental Perception

The simulated **LiDAR sensor**, implemented through ray-casting techniques, continuously scans the surrounding environment. Multiple sensor rays are projected around the vehicle to detect nearby traffic vehicles, pedestrians, road boundaries, and static obstacles. The measured distances are forwarded to the autonomous driving controller for decision-making.

4.3.3 Autonomous Decision Making

The autonomous controller analyses the planned route together with the real-time sensor information to determine the appropriate driving action. Based on surrounding traffic conditions, the controller performs:

- Steering adjustment

- Speed regulation
- Obstacle avoidance
- Collision prevention
- Safe turning at intersections
- Traffic signal compliance

These decisions are continuously updated throughout the navigation process.

4.3.4 Traffic and Pedestrian Management

Simultaneously, the traffic management module controls the movement of surrounding vehicles, while the pedestrian management module updates pedestrian movement across designated crossings. These dynamic entities create realistic traffic scenarios that continuously influence the autonomous driving behaviour.

4.4. Phase 4: Telemetry Collection and Performance Analysis

Throughout the simulation, the telemetry subsystem continuously records operational parameters generated by the autonomous vehicle. The recorded information includes:

- Vehicle speed
- Steering angle
- Vehicle position
- Travelled distance
- Acceleration
- Braking events
- Collision count
- Route completion status
- Sensor observations

- Safety-related performance indicators

These telemetry data are transmitted to the **Flask backend**, where they are processed and stored for performance evaluation. The collected information enables users to analyse vehicle behaviour, evaluate navigation efficiency, identify potential improvements, and compare autonomous driving performance under different traffic scenarios. Once the destination is reached, the simulation terminates successfully and the recorded performance data become available for further analysis and future research.

5. Use Case Diagram

The **Use Case Diagram** illustrates the functional interaction between the external actors and the **SmartDrive – 3D Self-Driving Car Simulator**. It provides a high-level view of the system by identifying the primary users and the services offered by the simulator. The diagram helps in understanding how different actors communicate with the system to perform autonomous driving operations and monitor simulation results.

The SmartDrive simulator involves three primary actors: **User/Tester**, **Autonomous Vehicle**, and the **Traffic Management System**. The **User/Tester** initiates the simulation by selecting the source and destination, enabling or disabling the autonomous driving mode, manually overriding the vehicle when necessary, and viewing the generated telemetry reports. The **Autonomous Vehicle** performs the core autonomous driving operations such as route generation, autonomous navigation, obstacle detection using the LiDAR sensor, traffic signal compliance, wrong-way recovery, pedestrian detection, and telemetry collection. The **Traffic Management System** controls the movement of non-player vehicles (NPCs) and traffic signals, creating realistic traffic scenarios that influence the autonomous vehicle's navigation decisions.

The interactions represented in the Use Case Diagram demonstrate the complete workflow of the SmartDrive simulator, beginning with user input and ending with performance evaluation after the successful completion of the journey. This diagram provides a conceptual understanding of the overall functionality of the proposed system before discussing its detailed architecture and implementation in the subsequent chapters.

Furthermore, the Use Case Diagram serves as a blueprint for identifying the functional boundaries of the SmartDrive simulator and clarifies the responsibilities of each actor involved in the system. It ensures that every major feature, including route planning, autonomous navigation, obstacle detection, traffic management, and telemetry analysis, is properly represented during the system design phase. The diagram also helps developers understand the sequence of interactions required to achieve successful autonomous navigation and provides a clear foundation for system implementation. By defining the relationships between the actors and the system functionalities, the Use Case Diagram simplifies requirement analysis, supports software development, and facilitates future enhancements.

Table 1.1: Description of Use Cases

Use Case	Description
Select Source & Destination	Allows the user to choose the starting point and destination for autonomous navigation.
Start / Stop Autonomous Mode	Enables or disables the autonomous driving mode during simulation.
Manual Driving Override	Allows the user to take manual control of the vehicle whenever required.
Generate Route using A*	Computes the shortest and most efficient path between the selected locations.
Autonomous Navigation	Controls the vehicle automatically while following the planned route.
Obstacle Detection using LiDAR	Detects nearby vehicles and obstacles using the simulated LiDAR sensor.
Traffic Signal Compliance	Ensures the autonomous vehicle obeys traffic lights and intersection rules.
Pedestrian Detection	Detects pedestrians and performs appropriate speed reduction or braking.
Wrong-Way Recovery	Detects incorrect vehicle movement and redirects it towards the correct route.
Telemetry Collection	Records operational parameters such as speed, steering angle, braking, and travelled distance.
Generate Performance Reports	Displays and stores the simulation results for performance evaluation and analysis.

Table2: Description of Use Case

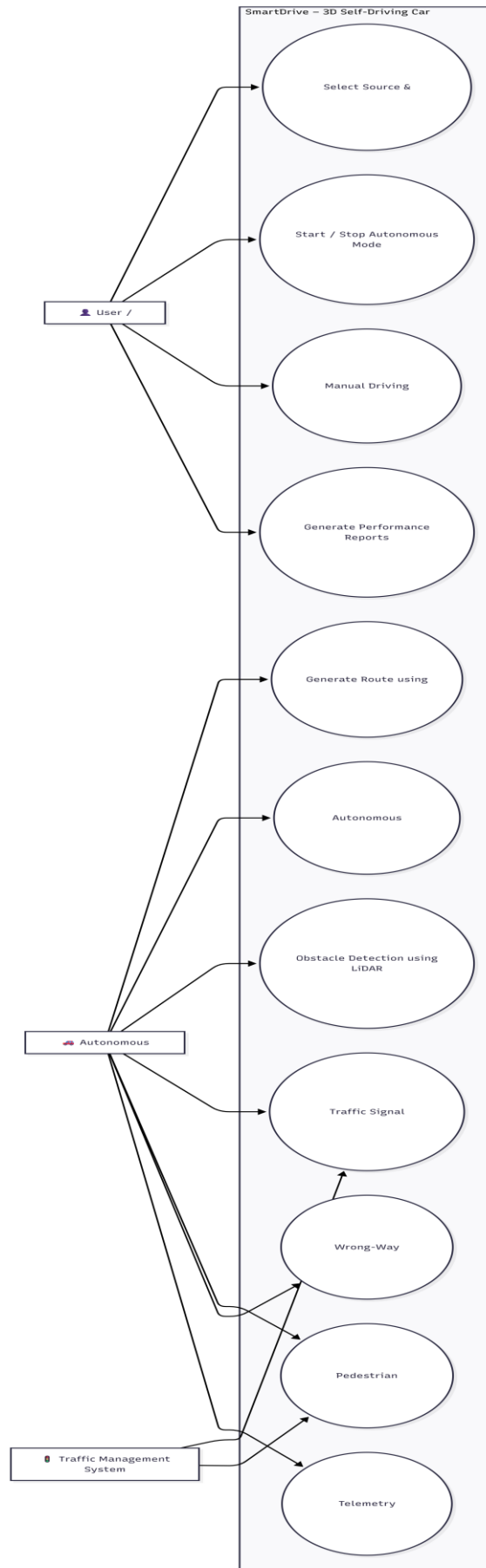


Fig2: Use Case Diagram

6. Data Flow Diagram

6.1. Level 0 Data Flow Diagram

The **Level 0 Data Flow Diagram (Context Diagram)** provides a high-level overview of the SmartDrive – 3D Self-Driving Car Simulator. It illustrates the interaction between the **User** and the **SmartDrive System**. The user provides the source, destination, and simulation controls as input. The simulator processes this information and returns the navigation results, vehicle status, and telemetry data. This diagram represents the overall flow of information

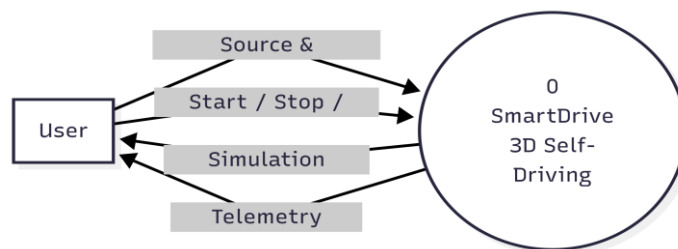


Fig3: Level 0 Data Flow Diagram

6.2. Level 1 Data Flow Diagram

The **Level 1 Data Flow Diagram** illustrates the internal data flow within the SmartDrive simulator. The user inputs are processed to generate the shortest route using the A* algorithm. The generated waypoints are sent to the vehicle controller, which receives data from the LiDAR and Traffic Management modules to perform autonomous navigation.

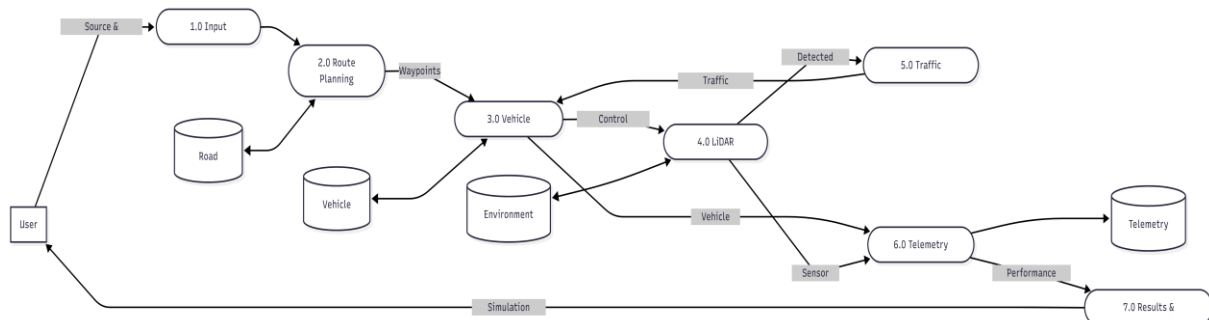


Fig4: Level 1 Data Flow Diagram

7. Proposed System Algorithm

7.1 Introduction

The SmartDrive – 3D Self-Driving Car Simulator integrates multiple algorithms to enable intelligent autonomous navigation within a three-dimensional virtual environment. Instead of relying on a single algorithm, the proposed system combines route planning, environmental perception, vehicle control, obstacle avoidance, traffic management, and telemetry collection into a unified autonomous driving framework. Each algorithm performs a specific task while continuously exchanging information with other system modules to ensure safe, efficient, and realistic vehicle navigation.

The execution of the autonomous driving system begins with route generation using the **A*** (A-Star) path-planning algorithm [2], [7]. The generated route is converted into navigation waypoints, which guide the autonomous vehicle throughout the simulation. During navigation, the simulated LiDAR sensor [8] continuously scans the environment for surrounding vehicles, pedestrians, traffic signals, and obstacles. The autonomous controller processes this information and generates steering, acceleration, and braking commands while ensuring compliance with traffic rules and collision avoidance. Simultaneously, telemetry data are recorded and transmitted to the backend for performance evaluation.

7.2. Assumptions

The proposed **SmartDrive – 3D Self-Driving Car Simulator** is developed based on the following assumptions:

- The simulation operates within a predefined urban road network containing roads, intersections, traffic signals, and parking areas.
- The user provides valid source and destination locations before starting the simulation.
- The **A*** algorithm generates the shortest feasible route between the selected locations.
- The simulated **LiDAR sensor**[8] accurately detects nearby vehicles, pedestrians, and obstacles within its sensing range.

- Traffic signals and NPC vehicles follow predefined behaviours throughout the simulation.
- The autonomous vehicle obeys traffic rules, performs obstacle avoidance, and follows the planned navigation path.
- Communication between the **Three.js frontend** and the **Flask backend** [5] remains stable during execution.
- The telemetry module accurately records vehicle performance and simulation data for analysis.

7.3 Overall Algorithm

Algorithm: SmartDrive Autonomous Navigation

Input

- Source Location
- Destination Location
- Road Network
- Traffic Information
- Sensor Inputs

Output:

- Safe Autonomous Navigation
- Telemetry Data
- Performance Analysis

Steps

1. Start the simulator and initialize all system modules.
2. Load the 3D road network and simulation environment.
3. Spawn the autonomous vehicle at the selected source location.
4. Initialize surrounding traffic vehicles and pedestrians.
5. Receive the source and destination selected by the user.
6. Execute the A* algorithm to calculate the shortest route.
7. Convert the generated route into navigation waypoints.
8. Activate the autonomous driving controller.
9. Continuously scan the environment using the simulated LiDAR sensor.
10. Detect nearby vehicles, pedestrians, road boundaries, and traffic signals.
11. Evaluate the surrounding traffic conditions.

12. Calculate steering angle, acceleration, and braking commands.
13. Update the vehicle position according to the generated control commands.
14. Repeat the sensing and control cycle until the destination is reached.
15. Record telemetry information during each simulation step.
16. Generate performance statistics after successful navigation.
17. Stop the simulation.

7.4. Pseudocode

```
BEGIN
Initialize Simulator
Load Road Network
Load Traffic System
Spawn Vehicle
User selects Source and Destination
Route ← A* Path Planning (Source, Destination)
Generate Waypoints (Route)
WHILE Destination not reached DO
    Scan Environment using LiDAR
    Detect Obstacles
    Detect Traffic Lights
    Detect Pedestrians
    IF Obstacle Found THEN
        Reduce Speed
    ELSE
        Continue Navigation
    END IF
    Calculate Steering
    Update Vehicle Position
    Store Telemetry Data
END WHILE
Generate Performance Report
END
```

7.5. Working of the Algorithm

The execution begins when the user selects the starting point and destination through the graphical interface. The Flask backend receives the request and applies the A* path-planning algorithm to determine the shortest route across the road network. The generated path is converted into multiple navigation waypoints, which are transmitted to the autonomous vehicle.

During every simulation frame, the vehicle continuously scans its surroundings using the simulated LiDAR sensor. The sensor measures the distance to nearby vehicles, pedestrians, traffic signals, and other obstacles. Based on these observations, the autonomous controller decides whether the vehicle should continue moving, reduce speed, stop, or change steering direction.

If an obstacle or pedestrian is detected within the safety range, the controller immediately reduces speed or applies braking. Similarly, when approaching a traffic signal, the vehicle obeys the current signal state before entering the intersection. The steering controller continuously updates the vehicle orientation so that it accurately follows the generated route.

Throughout the navigation process, telemetry information including speed, steering angle, travelled distance, braking events, collision count, and route completion status is continuously recorded and transmitted to the backend. After the vehicle reaches its destination, the collected data are analysed to evaluate navigation efficiency and overall system performance.

7.6. Backend Decision Making

The backend uses **Random Forest** and **Reinforcement Learning** techniques to improve autonomous driving decisions. The Random Forest model predicts suitable driving actions and steering corrections, while the Reinforcement Learning module continuously evaluates vehicle behaviour through rewards and penalties. Telemetry data are stored in log files for future analysis and model improvement.

7.7. Frontend Vehicle Control

The frontend controls vehicle movement using the **Pure Pursuit** steering algorithm. The steering angle is calculated based on the look-ahead waypoint, allowing the vehicle to smoothly follow the planned route. Vehicle speed is automatically reduced while negotiating sharp turns or approaching obstacles. If the vehicle moves away from the planned route, the recovery mechanism guides it back to the correct path.

7.8. Mathematical Models

The simulator uses the **Kinematic Bicycle Model** to calculate vehicle motion during steering and turning.

$$\frac{dx}{dt} = v \sin \theta$$

$$\frac{dz}{dt} = v \cos \theta$$

$$\frac{d\theta}{dt} = \frac{v}{L} \sin s$$

To generate smooth turns at intersections, the simulator applies the **Quadratic Bézier Curve**.

$$B(t) = (1 - t)^2 P_0 + 2(1 - t)t P_c + t^2 P_1$$

These mathematical models improve vehicle stability and ensure smooth autonomous navigation.

7.9. Advantages of the Proposed Algorithm

The proposed autonomous navigation algorithm offers several advantages:

- Generates the shortest navigation path using the A* algorithm.
- Supports real-time obstacle detection through LiDAR sensing.
- Performs intelligent steering and speed control.
- Maintains compliance with traffic signals and road rules.
- Reduces collision risk through continuous environmental monitoring.
- Records telemetry data for performance analysis.
- Provides a modular architecture for future AI enhancements.
- Suitable for both educational and research-oriented autonomous driving applications.

Note

This chapter gives an overview of the **complete algorithmic workflow**. In later chapters, you should explain the individual algorithms in detail, such as:

- **Road Network Module** – Road graph generation.
- **LiDAR Module** – Ray-casting algorithm.
- **Autonomous Driving Algorithm** – Decision-making logic.
- **Machine Learning Models** – Random Forest implementation.
- **Reinforcement Learning** – Q-Learning training and reward function.
- **Telemetry & Analytics** – Data collection and analysis.

8. Simulation Results

8.1. Result

8.1.1 Experimental Setup

The SmartDrive – 3D Self-Driving Car Simulator was evaluated under different driving scenarios to verify the performance of its autonomous navigation system. The simulation was conducted in a realistic three-dimensional urban environment consisting of roads, intersections, parking bays, traffic signals, zebra crossings, pedestrians, and non-player traffic vehicles (NPCs). The autonomous vehicle navigated between the selected source and destination using the A* path-planning algorithm while continuously monitoring its surroundings through the simulated LiDAR sensor.

To evaluate the reliability of the proposed system, multiple simulation runs were performed with different traffic densities. During each experiment, the telemetry module recorded important performance parameters such as vehicle speed, travelled distance, steering angle, braking events, collision count, and route completion status. The collected data were analysed to assess the efficiency, safety, and stability of the autonomous driving system.

Parameter	Symbol	Value	Unit
Cruise Speed	Vcruise	10.0	m/s
Maximum Speed	Vmax	22.0	m/s
Maximum Steering Angle	$\theta_{\max}$	0.6	rad
Vehicle Wheelbase	L	2.8	m
LiDAR Detection Range	Rlidar	50.0	m
Safety Braking Distance	Dsafe	12.0	m

Table 3: Simulator Parameters

8.1.2. Experimental Results

The experimental results demonstrate that the proposed **SmartDrive – 3D Self-Driving Car Simulator** successfully performs autonomous navigation under different traffic conditions. The vehicle generated the shortest route, safely followed the planned path, and responded effectively to nearby vehicles, pedestrians, and obstacles using the LiDAR sensor. As traffic density increased, the controller automatically adjusted the vehicle's speed to maintain safe navigation. The telemetry data confirmed accurate path following, effective obstacle avoidance, and stable overall system performance.

Performance Metric	Low Traffic	Medium Traffic	High Traffic
Route Successfully Generated	Yes	Yes	Yes
Autonomous Navigation	Successful	Successful	Successful
Obstacle Detection	Accurate	Accurate	Accurate
Traffic Signal Compliance	Yes	Yes	Yes
Pedestrian Detection	Successful	Successful	Successful
Collision Avoidance	Successful	Successful	Minor Delay
Route Completion	100%	100%	100%
Telemetry Recording	Successful	Successful	Successful

Table 4: Experimental Result

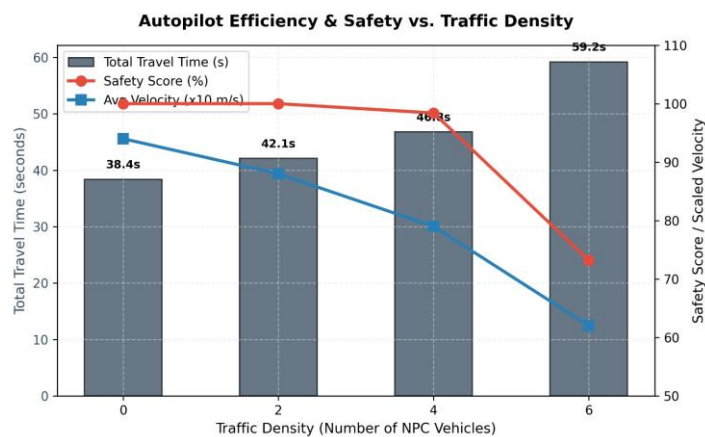


Fig 5: Autopilot Efficiency and Safety under Different

Figure 5: shows the performance of the SmartDrive simulator under different traffic densities. As the number of NPC vehicles increased, the total travel time gradually increased while the average vehicle speed decreased. The safety score remained high under low and moderate traffic conditions but dropped slightly in heavy traffic due to increased obstacle interactions. These results indicate that the autonomous controller effectively adapts its speed and navigation strategy to maintain safe and efficient driving.

8.1.3 Comparative Analysis

The proposed SmartDrive – 3D Self-Driving Car Simulator offers several improvements over traditional autonomous driving simulators by integrating intelligent route planning, LiDAR-based obstacle detection, realistic vehicle physics, traffic management, and telemetry analysis within a modular web-based architecture. Unlike conventional systems that provide limited functionality, SmartDrive enables safe autonomous navigation in a dynamic 3D environment while supporting future enhancements.

Feature	Traditional Simulator	SmartDrive Simulator
Route Planning	Static/Basic	A* Path Planning
Obstacle Detection	Limited	LiDAR-based Detection
Traffic Management	Basic Traffic	Dynamic Traffic System
Pedestrian Detection	Not Available	Supported
Telemetry	Limited Analysis	Real-time Telemetry
Architecture	Monolithic	Modular Client-Server

Table 5: Comparative Analysis of Traditional Simulator and Smart Drive Simulator

8.1.4. Performance Analysis

The obtained results confirm that the integration of the **A*** path-planning algorithm, simulated **LiDAR** sensing, realistic vehicle dynamics, and autonomous decision-making enables safe and reliable navigation within a dynamic urban environment. The generated routes remained efficient while the autonomous controller continuously adapted to changing traffic conditions by regulating speed and steering.

The telemetry analysis further indicates that the vehicle maintained stable navigation performance, complied with traffic signals, and successfully avoided obstacles during most simulation scenarios. Although higher traffic density increased travel time and reduced the average vehicle speed, the autonomous controller continued to operate effectively by dynamically adjusting its driving behaviour. Overall, the proposed SmartDrive simulator provides a realistic and efficient platform for studying autonomous driving technologies.

Test Case	Expected Result	Observed Result	Status
System Initialization	Simulator starts successfully	Successful	Pass
Route Generation	Shortest path generated	Successful	Pass
Vehicle Navigation	Vehicle follows planned route	Successful	Pass
LiDAR Detection	Obstacles detected	Successful	Pass
Traffic Signal Detection	Vehicle obeys traffic lights	Successful	Pass
Pedestrian Detection	Vehicle slows or stops	Successful	Pass
Telemetry Recording	Performance data stored	Successful	Pass
Destination Reached	Vehicle reaches destination	Successful	Pass

Table 6: Functional Testing Result

Parameter	Observation
Navigation Accuracy	High
Obstacle Detection	Reliable
Traffic Rule Compliance	Successful
Route Planning Efficiency	High
Vehicle Stability	Stable
Telemetry Generation	Accurate
Overall System Performance	Satisfactory

Table 7: Performance Summary

8.2. System Screenshots

The following screenshots present the successful implementation and execution of the SmartDrive – 3D Self-Driving Car Simulator. They demonstrate the major functionalities of the proposed system, including autonomous navigation, route generation, obstacle detection, traffic management, and telemetry analysis.

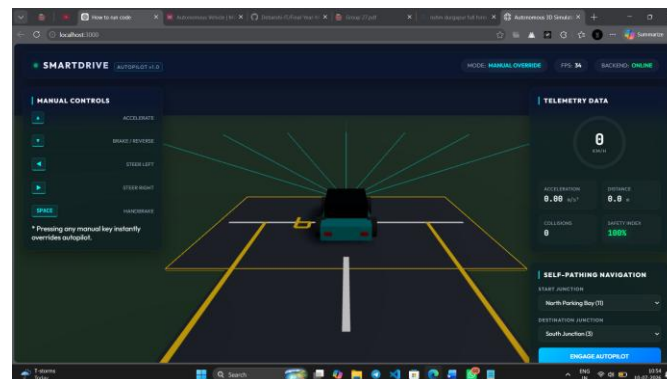


Fig 6: Home Screen of SmartDrive Simulator

Figure 5: illustrates the main interface of the SmartDrive – 3D Self-Driving Car Simulator. The interface consists of the Manual Controls panel, Telemetry Dashboard, and Self-Pathing Navigation module. From this screen, the user can configure the route, monitor vehicle performance, and switch between manual and autonomous driving modes.

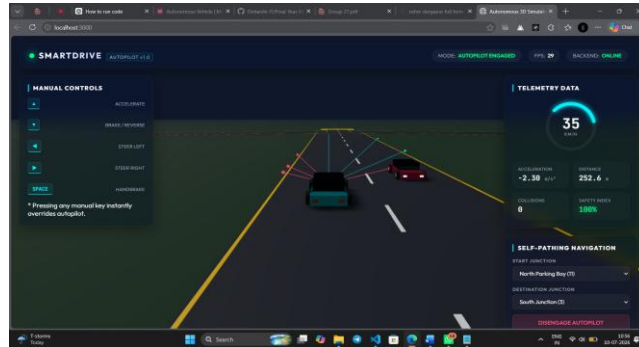


Fig 7: Route Selection and Autopilot Activation

Figure 6: shows the route selection interface after the start and destination junctions have been selected. The user can engage the autonomous driving mode using the **Engage Autopilot** button, after which the simulator automatically begins route execution.

9. Conclusion and Future Work

9.1 Conclusion

The SmartDrive – 3D Self-Driving Car Simulator was successfully designed and developed as a web-based autonomous driving simulation platform that demonstrates the core principles of intelligent vehicle navigation in a realistic three-dimensional environment. The system integrates Three.js for 3D visualization, Python Flask for backend communication, the A* path-planning algorithm for route generation, simulated LiDAR sensing for obstacle detection, realistic vehicle physics, traffic management, and telemetry analysis.

The simulator enables autonomous navigation by generating the shortest path, detecting obstacles, obeying traffic signals, avoiding collisions, and safely reaching the selected destination. The modular architecture ensures efficient communication between different system components and allows future enhancements without affecting the existing implementation. The project successfully meets its objectives by providing a practical, safe, and cost-effective platform for learning and evaluating autonomous driving technologies. It also serves as an educational tool for understanding Artificial Intelligence, Intelligent Transportation Systems, and autonomous vehicle development.

9.2 Future Work

The proposed **SmartDrive – 3D Self-Driving Car Simulator** provides a strong foundation for autonomous driving research and can be further enhanced in several ways. Future improvements include integrating **Computer Vision** for lane and traffic sign detection, implementing advanced **Deep Learning** and **Reinforcement Learning** algorithms, supporting additional sensors such as Radar and GPS, and simulating different weather and road conditions. The simulator can also be extended with **Vehicle-to-Vehicle (V2V)** communication, larger road networks, and improved telemetry analysis. These enhancements will increase the realism, intelligence, and research capabilities of the proposed system.

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